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Paint roller assembly

Abstract

A paint roller assembly for allowing a sufficient amount of pressure to be applied for distributing paint evenly is disclosed. The paint roller assembly includes a pair of frame members. One frame member includes a handle. A cylindrical-shaped core is attached to another end of each of the frame members. In one aspect, the pair of frame members are coupled to each other at a first end by a variety of different coupling mechanisms. When assembled, a central, longitudinal axis of the handle is substantially parallel to a central, longitudinal axis of the cylindrical-shaped core, thereby allowing the user to apply sufficient pressure to more evenly distribute paint during use.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The invention pertains to the field of paint rollers. More particularly, the invention pertains to a paint roller assembly with a handle that is substantially parallel to the roller or paint applicator.

BACKGROUND OF THE DISCLOSURE

(2) A traditional roller brush consists of a wire frame and a handle pointing straight at the roller. With a traditional roller brush, your hand is usually flexed forward, which puts stress on your wrist and forearm. In addition, due to the flimsy construction of the wire frame, you also find yourself working harder to apply paint in a consistent manner. Normally, you have to push harder on the end of the roller that doesn't have the frame connected. This often leaves a line of paint on the other side of the roller.

(3) In view of the foregoing, it would be desirable to provide a paint roller assembly that overcomes the above-mentioned shortcomings of a conventional roller brush.

SUMMARY OF THE DISCLOSURE

(4) While painting my back deck, I came up with an idea for a new frame that would help eliminate the fatigue associated with painting for extended periods of time. With my paint roller assembly, the handle runs parallel to the roller cover or paint applicator, which allows you to hold the assembly with a natural, comfortable hand position. By doing so, the invention helps to eliminate the wrist and forearm fatigue associated with using conventional paint roller assemblies for extended periods of time. Because the support members or frame members support the roller cover from both sides of the roller cover, the paint roller assembly of the invention also provides a more consistent finished product. The solid construction and design of the handle allow you to apply force to the paint roller assembly, and in particular, to the roller cover or paint applicator, with your entire arm and in many cases, your whole body.

(5) In one aspect of the invention, a paint roller assembly comprises a pair of frame members. A handle is attached to a first end of each of the frame members. A cylindrical-shaped core is attached to a second end of each of the frame members. The cylindrical-shaped core is configured to rotate about a central, longitudinal axis, wherein the central, longitudinal axis of the handle is substantially parallel to a central, longitudinal axis of the cylindrical-shaped core when the paint roller assembly is assembled. The parallel arrangement between the handle and the cylindrical-shaped core enables sufficient pressure to be applied to more evenly distribute paint during use of the paint roller assembly.

(6) In another aspect of the invention, a paint roller assembly comprises a first part and a second part coupled to each other at a first end by a coupling mechanism. The second part includes a handle (having a central, longitudinal axis. A cylindrical-shaped core is attached to a second end of each of the first part and the second part. The cylindrical core is configured to rotate about a central, longitudinal axis. The central, longitudinal axis of the handle is substantially parallel to a central, longitudinal axis of the cylindrical-shaped core when the paint roller assembly is assembled. Similar to the earlier aspect of the invention, the parallel arrangement between the handle and the cylindrical-shaped core enables sufficient pressure to be applied to more evenly distribute paint during use of the paint roller assembly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) While various embodiments of the invention are illustrated, the particular embodiments shown should not be construed to limit the claims. It is anticipated that various changes and modifications

may be made without departing from the scope of this invention.
(2) FIG. **1**A is a front perspective view of a paint roller assembly when assembled according to an aspect of the invention;
(3) FIG. **1**B is an exploded view of the paint roller assembly of FIG. **1**A when assembled;
(4) FIG. **2** is a front view of a paint roller assembly of FIG. **1**A including a frame member with a hinge according to an aspect of the invention;
(5) FIG. **3**A is an exploded perspective view of a paint roller assembly with an ergonomic handle and two parts that are coupled together using a screw-on coupling mechanism according to another aspect of the invention;
(6) FIG. **3**B is a perspective view of a paint roller assembly of FIG. **3**A when assembled;
(7) FIG. **4** is an exploded perspective view of a paint roller assembly with two parts that are coupled together using a twist-n-lock coupling mechanism according to another aspect of the invention;
(8) FIG. **5** is an exploded perspective view of a paint roller assembly with two parts that are coupled together using a pinch-n-release coupling mechanism according to another aspect of the invention;
(9) FIG. **6** is an exploded perspective view of a paint roller assembly including an ergonomic handle that is coupled together using a slide-n-lock coupling mechanism according to another aspect of the invention; and
(10) FIG. **7** is an exploded view of a paint roller assembly of FIG. **3**A that includes an extension kit according to another aspect of the invention.

DETAILED DESCRIPTION

(11) Directional phrases used herein, such as, for example, left, right, front, back, top, bottom and derivatives thereof, relate to the orientation of the elements shown in the drawings and are not limiting upon the claims unless expressly recited therein. Identical parts are provided with the same reference number in all drawings.
(12) Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about”, “approximately”, and “substantially”, are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value. Here and throughout the specification and claims, range limitations may be combined and/or interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise.
(13) Throughout the text and the claims, use of the word “about” in relation to a range of values (e.g., “about 22 to 35 wt %”) is intended to modify both the high and low values recited, and reflects the penumbra of variation associated with measurement, significant figures, and interchangeability, all as understood by a person having ordinary skill in the art to which this disclosure pertains.
(14) For purposes of this specification (other than in the operating examples), unless otherwise indicated, all numbers expressing quantities and ranges of ingredients, process conditions, etc., are to be understood as modified in all instances by the term “about”. Accordingly, unless indicated to the contrary, the numerical parameters set forth in this specification and attached claims are approximations that can vary depending upon the desired results sought to be obtained by embodiments. At the very least, and not as an attempt to limit the application of the doctrine of equivalents to the scope of the claims, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. Further, as used in this specification and the appended claims, the singular forms “a”, “an” and “the” are intended to include plural referents, unless expressly and unequivocally limited to one referent.

(15) Notwithstanding that the numerical ranges and parameters setting forth the broad scope are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from the standard deviation found in their respective testing measurements including that found in the measuring instrument. Also, it should be understood that any numerical range recited herein is intended to include all sub-ranges subsumed therein. For example, a range of “1 to 10” is intended to include all sub-ranges between and including the recited minimum value of 1 and the recited maximum value of 10, i.e., a range having a minimum value equal to or greater than 1 and a maximum value of equal to or less than 10. Because the disclosed numerical ranges are continuous, they include every value between the minimum and maximum values. Unless expressly indicated otherwise, the various numerical ranges specified in this application are approximations.
(16) The singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise.
(17) As shown in FIGS. **1**A and **1**B, a paint roller assembly **100** is shown according to one aspect of the invention. The paint roller assembly **100** has a cylindrical-shaped handle **114** and a pair of opposing frame members or support arms **112**. The frame members **112** are preferably made of suitable material, such as metal, and the like. A cylindrical-shaped core **116** comprising a spring cage **130** and an end cap **128** is mounted or attached to one end of each frame member or support arm **112**, which is inserted into a roller cover or paint applicator **132**. The spring cage **130** and the end cap **128** are made of a suitable material, such as plastic, and the like. The opposite end of each frame member or support arm **112** is inserted into each side of the handle **114** and into the end caps **128**, respectively. Once the paint roller assembly **100** is assembled, a central, longitudinal axis **118** of the handle **114** is substantially parallel to a central, longitudinal axis **126** of the cylindrical-shaped core **116**. This parallel configuration allows the user to apply sufficient pressure to the paint roller assembly **100** to more evenly distribute paint during use as compared to conventional paint roller assemblies.
(18) Located in the middle of the handle **114**, on the top, is a threaded hole **120**. This threaded hole **120** is used to support a conventional handle **134** or an extension pole that can be threaded into the hole **120**. When not in use, a plug **124** can be used to seal the threaded hole **120** and prevent dirt, paint, etc. from entering the threaded hole **120**. Unfortunately, this embodiment of the invention is a fixed design, which doesn't allow the consumer to change out the roller cover **132**. The materials used to construct the paint roller assembly **100** can include, but are not limited to, plastic, wood, metal, etc. In addition, the size of all parts, including the main frame, handle, cylindrical core, etc. include, but are not limited to, the proportions shown in the illustrations.
(19) Referring now to FIG. **2**, a paint roller assembly **200** is shown according to another aspect of the invention. Similar to the paint roller assembly **100**, the paint roller assembly **200** includes a handle **214** made of suitable material, such as wood, and the like, and two frame members or support arms **202**, **204** made out of suitable material, such as metal, and the like. At the end of each support arm **202**, **204** is a cylindrical core **216**, which is inserted into the roller cover **232**. The cylindrical core **216** comprises a spring cage **230** and two end caps **228** made of suitable material, such as plastic, and the like. The opposite end of each support arm **202**, **204** is permanently attached to each side of the handle **214**. Unlike the paint roller assembly **100**, the paint roller assembly **200** has hinges **234** on each support arm **202**, **204**, which allow the arms **202**, **204** to open to remove or replace the roller cover **232**. Once the paint roller assembly **200** is assembled, a central, longitudinal axis **218** of the handle **214** is substantially parallel to a central, longitudinal axis **226** of the cylindrical-shaped core **216**. This parallel configuration allows the user to apply sufficient pressure to the paint roller assembly **200** to more evenly distribute paint during use as compared to conventional paint roller assemblies.
(20) Similar to the embodiment of FIG. **1**A, a threaded hole **220** is located on the top and in the middle of the handle **214**. This hole **220** is used to support a conventional handle, for example, the

conventional handle **134**, or an extension pole (not shown). When not in use, a plug **228** is used to seal the hole **220** and prevent dirt, paint, etc. from entering the opening **220**. The materials used to construct this paint roller assembly **200** can include, but are not limited to, plastic, wood, metal, etc. In addition, the size of all parts, including the main frame, handle, cylindrical core, etc. include, but are not limited to, the proportions shown in the illustrations.

(21) Referring now to FIGS. **3**A ad **3**B, a paint roller assembly **300** is shown according to another aspect of the invention. In this aspect, the paint roller assembly **300** includes two frame members or support arms **302**, **304** that are coupled together using a screw-on coupling mechanism **306**. One frame member **304** has an ergonomic handle **314** that includes a fingerhold **309** and a male threaded end cap **308**. In the illustrated embodiment, the fingerhold **307** supports the fingers of the user and provides a more comfortable or more secure hold on the ergonomic handle **314**. On the opposite end is a cylindrical core **316** that is inserted into a roller cover **332**. The cylindrical core **316** comprises a spring cage **330** and a plastic end cap **328**. The spring cage **330** and the end cap **328** are made of a suitable material, such as plastic, and the like. The other frame member **302** includes a female threaded coupler **310**. On the opposite end of the frame member **302** is another cylindrical core **316** that is inserted into the roller cover or paint applicator **332**. Once both end caps **328** have been inserted in the roller cover **332**, the two frame members **302**, **304** of the paint roller assembly **300** will be connected using the screw-on coupling mechanism **306**. To couple the two frame members **302**, **304** together, the user inserts the male end cap **308** into the female coupler **310** and turn or rotate the male end cap **308** until tightly coupled to the female coupler **310**. Once the paint roller assembly **300** is assembled in this manner, a central, longitudinal axis **312** of the handle **314** is substantially parallel to a central, longitudinal axis **326** of the cylindrical-shaped core **316**. This parallel configuration allows the user to apply sufficient pressure to the paint roller assembly **300** to more evenly distribute paint during use as compared to conventional paint roller assemblies.

(22) Located in the middle of the handle **314**, on the top, is a threaded hole **320**. This hole **320** is used to support a conventional handle **334** or extension pole (not shown). When not in use, a plug **324** is used to seal the hole **320** and prevent dirt, paint, and the like, from entering the opening. Because the paint roller assembly **300** is constructed of two parts that separate, the roller cover **332** can be removed or replaced at any time. The materials used to construct this paint roller assembly **300** can include, but are not limited to, plastic, wood, metal, etc. In addition, the size of all parts, including the main frame, handle, cylindrical core, etc. include, but are not limited to, the proportion shown in the illustrations.

(23) Referring now to FIG. **4**, a paint roller assembly **400** is shown according to another aspect of the invention. In this aspect, the paint roller assembly **400** includes two frame members or support arms **402**, **404** that are coupled together by a twist-n-lock coupling mechanism **406**. One frame member **404** has an ergonomic handle **414** that includes a fingerhold **407** and a male end cap **408** with one thread. Similar to the fingerhold **307**, the fingerhold **407** is designed to allow the user a more comfortable or more secure hold on the ergonomic handle **414**. On the opposite end is a cylindrical core **416** that is inserted into a roller cover or paint applicator **432**. The cylindrical core **416** comprises a spring cage **430** and an end cap **428**, preferably made of a suitable material, such as plastic, and the like. The other frame member **402** includes a female coupler **410**. On the opposite end of the other frame member **402** is another cylindrical core **416** that is inserted into the roller cover **432**. Once both ends have been inserted in the roller cover **432**, the two frame members **402**, **404** of the paint roller assembly **400** can be coupled together using the twist-n-lock coupling mechanism **406**. To couple the two frame members **402**, **404** of the paint roller assembly **400** together, the male end cap **408** is inserted into the female coupler **410**. Pressure is applied to both frame members **402**, **404**, causing both ends to squeeze together. The female coupler **410** is then rotate a half turn, at which point, the two ends **402**, **404** are locked. Once the paint roller assembly **400** is assembled, a central, longitudinal axis **418** of the handle **414** is substantially parallel to a

central, longitudinal axis **426** of the cylindrical-shaped core **416**. This parallel configuration allows the user to apply sufficient pressure to the paint roller assembly **400** to more evenly distribute paint during use as compared to conventional paint roller assemblies.

(24) Located in the middle of the handle **414**, on the top, is a threaded hole **420**. This hole **420** is used to receive a conventional handle **434** or extension pole. When not in use, a plug **424** is used to seal the hole **420** and prevent dirt, paint, etc. from entering the opening. Because this paint roller assembly **400** is constructed of two frame members **402**, **404** that separate, the roller cover **432** can be removed or replaced at any time. The materials used to construct this paint roller assembly can include, but are not limited to, plastic, wood, metal, etc. In addition, the size of all parts, including the main frame, handle, cylindrical core, etc. include, but are not limited to, the proportions shown in the illustrations.

(25) Referring now to FIG. **5**, a paint roller assembly **500** is shown according to another aspect of the invention. In this aspect, the paint roller assembly **500** includes two frame members or support arms **502**, **504** that are coupled together by a pinch-n-release coupling mechanism **506**. One frame member **504** has an ergonomic handle **514** that includes a fingerhold **507** and a female end cap **508**. Similar to the fingerholds **307**, **407**, the fingerhold **507** is designed to allow the user a more comfortable or more secure hold on the handle **514**. The female end cap **508** has two small, square shaped openings **509** on both sides of the handle **514**. On the opposite end of the frame member **504** is a cylindrical core **516** that is inserted into the roller cover **532**. The cylindrical core **516** comprises a spring cage **530** and an end cap **528**, preferably made of a suitable material, such as plastic, and the like. The other frame member **502** includes a male end cap **510** that has two protruding square tabs **511** on each side. On the opposite end of the other frame member **502** is another cylindrical core **516** that is inserted into the roller cover **534**. Once both ends of the frame members **502**, **504** have been inserted in the roller cover **532**, the two frame members **502**, **504** of the paint roller assembly **500** will be connected using the pinch-n-release coupling **506**. To attach the two frame members **502**, **504** of the paint roller assembly **500**, the male end cap **510** is inserted into the female end cap **508**. Pressure is applied to both frame members **502**, **504** until the two tabs **511** on the male end cap **510** are aligned with the openings **509** on the female end cap **508**. At this point, the two tabs **511** can be released and are received within the openings **509** to firmly lock frame members **502**, **504** in place. Once the paint roller assembly **500** is assembled, a central, longitudinal axis **518** of the handle **514** is substantially parallel to a central, longitudinal axis **526** of the cylindrical-shaped core **516**. This parallel configuration allows the user to apply sufficient pressure to the paint roller assembly **500** to more evenly distribute paint during use as compared to conventional paint roller assemblies.

(26) Located in the middle of the handle **514**, on the top, is a threaded hole **520**. The threaded hole **520** is used to support a conventional handle **534** or extension pole. When not in use, a plug **528** is used to seal the hole **520** and prevent dirt, paint, etc. from entering the opening. Because this paint roller assembly **400** is constructed of two frame members **502**, **504** that separate, the roller cover **532** can be removed or replaced at any time. The materials used to construct this paint roller assembly can include, but are not limited to, plastic, wood, metal, etc. In addition, the size of all parts, including the main frame, handle, cylindrical core, etc. include, but are not limited to, the proportions shown in the illustrations.

(27) Referring now to FIG. **6**, a paint roller assembly **600** is shown according to another aspect of the invention. In this aspect, the paint roller assembly **600** includes an ergonomic handle **614** that uses a slide-n-lock coupling mechanism **606** and two frame members or support arms **602**, **604** made out of a suitable material, such as metal, and the like. The ergonomic handle **614** includes a fingerhold **607**. Similar to the fingerholds **307**, **407**, **507**, the fingerhold **607** is designed to allow the user a more comfortable or more secure hold on the ergonomic handle **614**. The ergonomic handle **614** is split in the middle, forming two halves, an upper part **614***a* and lower part **614***b*. At the end of each arm **602**, **604** is a cylindrical core **616**, which is inserted into the roller cover **632**.

The cylindrical core **616** comprises a spring cage **630** and two plastic end caps **628**. The opposite end of each metal frame member or support arm **602**, **604** is bent to form an eye loop or circle. Once both cylindrical cores **616** are inserted into the roller cover **632**, the eye loops on each arm **602**, **604** are locked in grooves **614***c* located in the lower half of the handle. The upper and lower handles are then placed together, in a slightly offset position. They are then slid together, so they line up, forming the handle of the paint roller assembly. Once the paint roller assembly **600** is assembled, a central, longitudinal axis **618** of the handle **614** is substantially parallel to a central, longitudinal axis **626** of the cylindrical-shaped core **616**. This parallel configuration allows the user to apply sufficient pressure to the paint roller assembly **600** to more evenly distribute paint during use as compared to conventional paint roller assemblies.
(28) Located in the middle of the handle, on the top, is a threaded hole **620**. This hole **620** can be used to support a conventional handle **634** or an extension pole. When not in use, a plug **628** is used to seal the hole **620** to prevent dirt, paint, etc. from entering the opening **620**. The materials used to construct this paint roller assembly **600** can include, but are not limited to, plastic, wood, metal, etc. In addition, the size of all parts, including the main frame, handle, cylindrical core, etc. include, but are not limited to, the proportions shown in the illustrations.
(29) Referring now to FIG. **7**, a paint roller assembly extension kit **722** is shown according to an aspect of the invention. The extension kit **722** can be used with any of the earlier embodiments of the paint roller assemblies described above. In the illustrated embodiment, for example, the paint roller assembly **300** with the screw-on coupling mechanism **306** is shown. The extension kit **722** can be constructed in a manner that allows the paint roller assembly to be expanded. The extension kit **722** can be purchased separately to expand the size (i.e., the width) of the roller cover or paint applicator **732**. In one aspect, the extension kit **722** can easily double the size (i.e., width) of a traditional 9-inch roller cover or paint applicator to at least 18 inches or more. Unfortunately, due to the fact that a rod **724** made of suitable material, such as plastic, and the like, is inserted in the frame members **302**, **304** for support, the extension kit **722** can only be used with a conventional handle **734**. This conventional handle **734** is permanently attached to an extension piece **726**, in what will be the center of the new handle, once it has been connected to the paint roller assembly **300**. Depending on the actual handle embodiment, the extension will use the same method for attaching to the paint roller assembly **300**. In addition, the conventional handle **734** will be threaded on the end to allow an extension pole as in all the earlier embodiments, thereby providing greater reach for those hard-to-reach spots, for example, a ceiling, a floor, and the like.
(30) Accordingly, it is to be understood that the embodiments of the invention herein described are merely illustrative of the application of the principles of the invention. Reference herein to details of the illustrated embodiments is not intended to limit the scope of the claims, which themselves recite those features regarded as essential to the invention.

Claims

1. A paint roller assembly, comprising: a pair of frame members; a coupling mechanism for coupling the pair of frame members to each other; an ergonomic handle attached to a first end of one of the frame members, the ergonomic handle having a fingerhold and a central, longitudinal axis; and a cylindrical-shaped core attached to a second end of each of the frame members, the cylindrical-shaped core configured to rotate about a central, longitudinal axis, wherein the central, longitudinal axis of the ergonomic handle is substantially parallel to the central, longitudinal axis of the cylindrical-shaped core when the paint roller assembly is assembled, thereby enabling a user to grasp the ergonomic handle and apply pressure to the paint roller assembly to evenly distribute paint during use.
2. The paint roller assembly of claim 1, further comprising a roller cover removably attached to each cylindrical-shaped core for collecting and dispersing paint.

3. The paint roller assembly of claim 1, wherein the coupling mechanism comprises a screw-on coupling mechanism in which the ergonomic handle includes a male threaded end cap on one frame member that is received within a female threaded coupler on the other frame member.
4. The paint roller assembly of claim 1, wherein the coupling mechanism comprises a twist-n-lock coupling mechanism in which the ergonomic handle includes a male end cap on one frame member that is received within a female coupler on the other frame member.
5. The paint roller assembly of claim 1, wherein the coupling mechanism comprises a pinch-n-release coupling mechanism in which the ergonomic handle includes a female end cap on one frame member that is received within a male end cap on the other frame member.
6. The paint roller assembly of claim 1, wherein the coupling mechanism comprises a slide-n-lock coupling mechanism in which the ergonomic handle includes two halves that is received within eye loops on each frame member.
7. The paint roller assembly of claim 1, wherein at least one frame member includes a hinge for allowing the at least one frame member to open and enable removal or replacement of a roller cover.
8. The paint roller assembly of claim 1, wherein the ergonomic handle includes a threaded hole or receiving a conventional handle.
9. The paint roller assembly of claim 1, wherein the cylindrical-shaped core comprises a spring cage and an end cap.
10. A paint roller assembly, comprising: a first frame member and a second frame member coupled to each other by a coupling mechanism, the second frame member including an ergonomic handle having a fingerhold and a central, longitudinal axis; and a cylindrical-shaped core attached to a second end of each of the first frame member and the second frame member, the cylindrical core configured to rotate about a central, longitudinal axis, wherein the central, longitudinal axis of the ergonomic handle is substantially parallel to the central, longitudinal axis of the cylindrical-shaped core when the paint roller assembly is assembled, thereby enabling a user to grasp the ergonomic handle and apply pressure to the paint roller assembly to evenly distribute paint during use.
11. The paint roller assembly of claim 10, wherein the coupling mechanism comprises a screw-on coupling mechanism in which the ergonomic handle includes a male threaded end cap that is received within a female threaded coupler on the other frame member.
12. The paint roller assembly of claim 10, wherein the coupling mechanism comprises a twist-n-lock coupling mechanism in which the ergonomic handle includes a male end cap that is received within a female coupler on the other frame member.
13. The paint roller assembly of claim 10, wherein the coupling mechanism comprises a pinch-n-release mechanism in which the ergonomic handle includes a female end cap that is received within a male end cap on the other frame member.
14. The paint roller assembly of claim 10, further comprising an extension kit for allowing the paint roller assembly to be expanded in width.
15. The paint roller assembly of claim 10, further comprising a roller cover removably attached to each cylindrical-shaped core for collecting and dispersing paint.